

Klasyfikacja retinopatii cukrzycowej na podstawie zdjęć dna oka

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Dataset



Trening i walidacja na podstawie zdjęć opartych na stadiach choroby (0 – zdrowy, 1 – 4 – różne stadia)
Trening: 6260 zdjęć dna oka
Walidacja: 2503 zdjęć dna oka

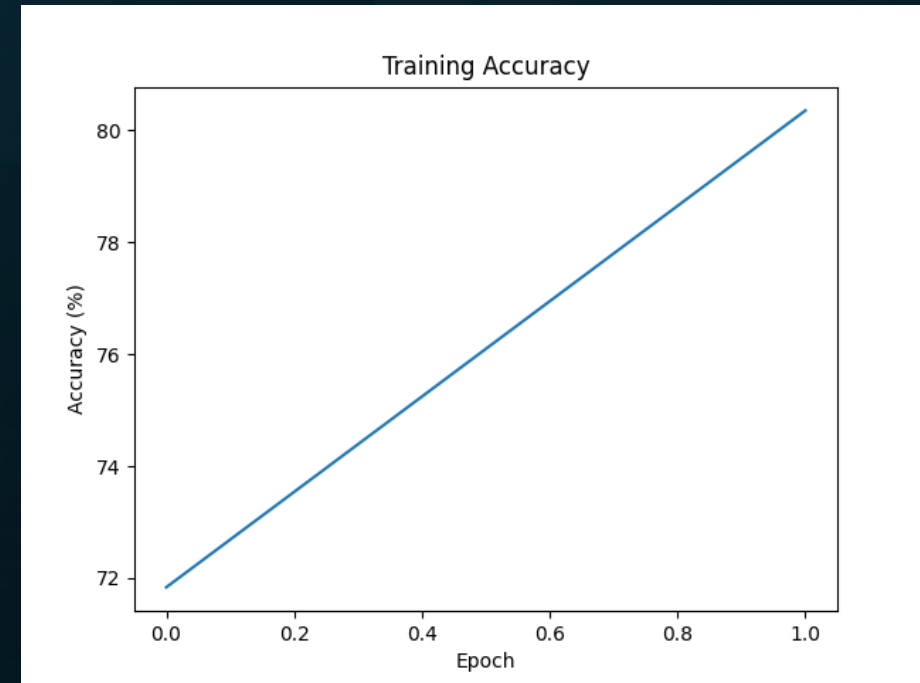
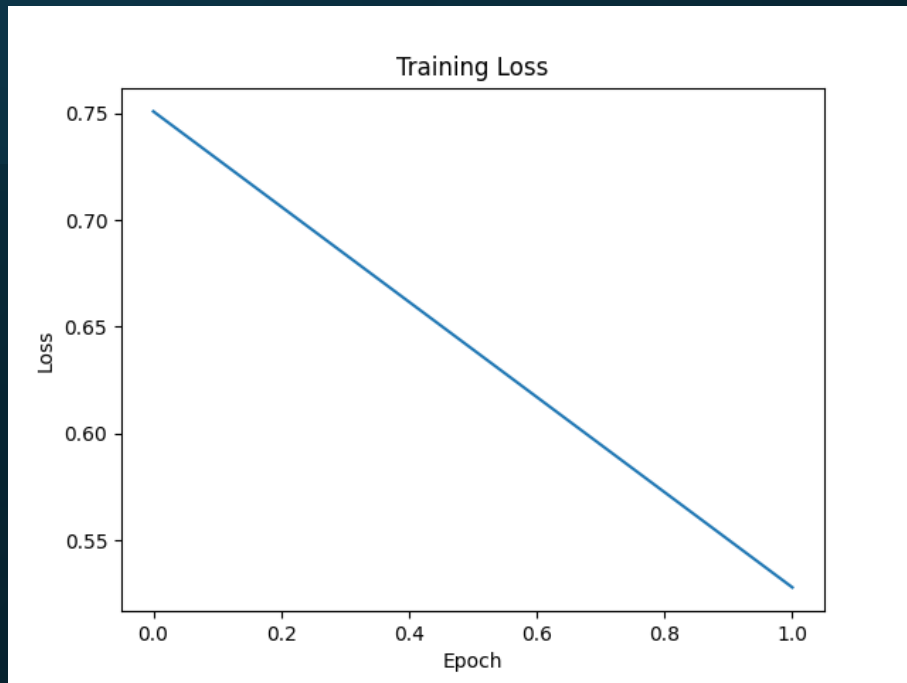


Test na podstawie zbiorów 0 – zdrowy 1- chory
Test: 243 zdjęć dna oka

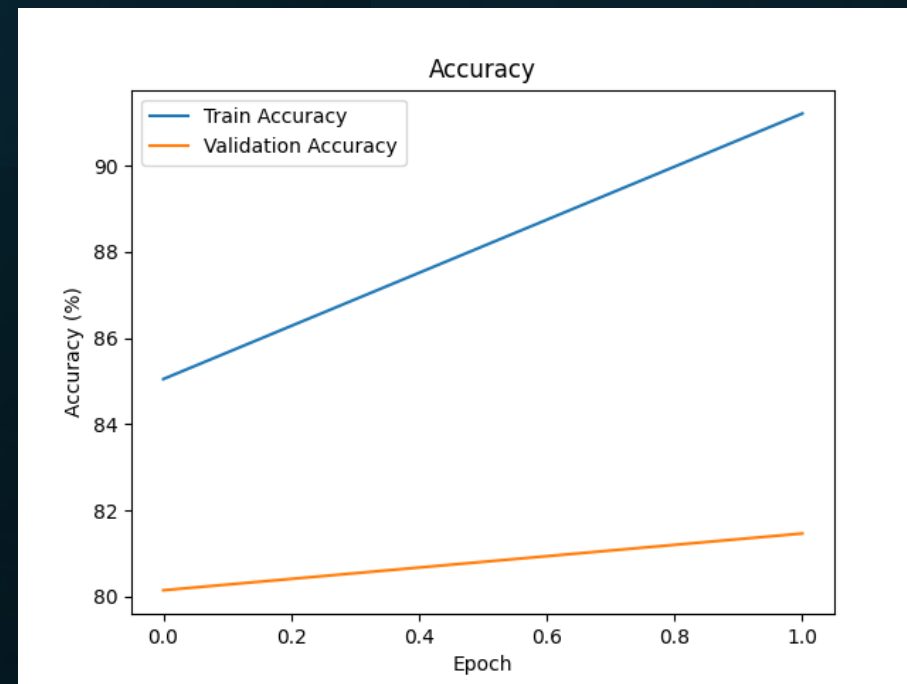
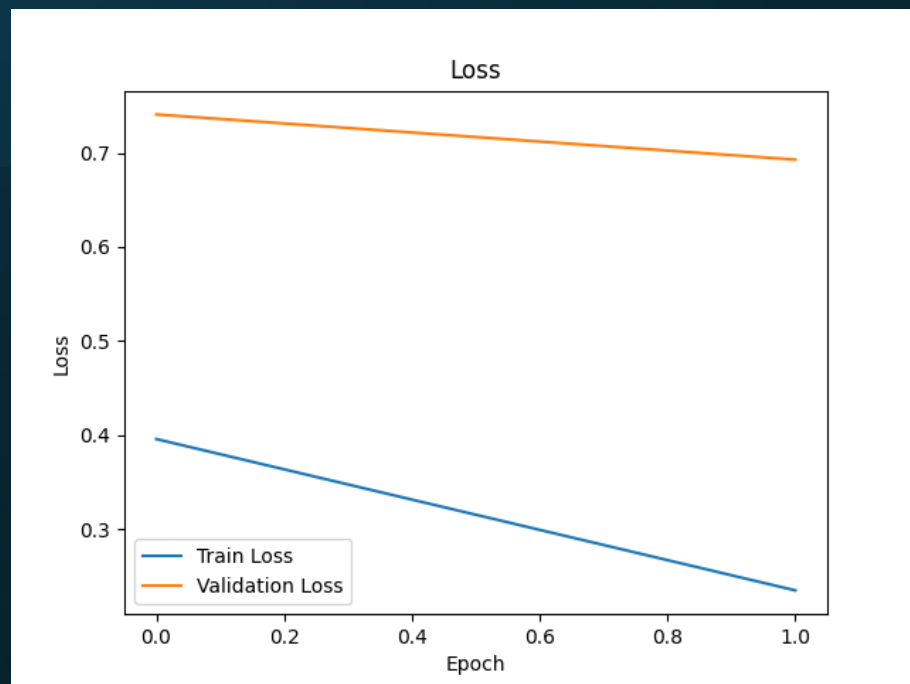
Forma treningu modelu

- Model: ResNet-18 (analiza obrazów medycznych)
- Transfer learning (wagi wstępnie wytrenowane na ImageNet)
- Resize obrazów do 224×224 px
- Konwersja obrazów do tensorów
- Trening w seriach po 2 epoki
- Zapisywanie modelu między kolejnymi treningami

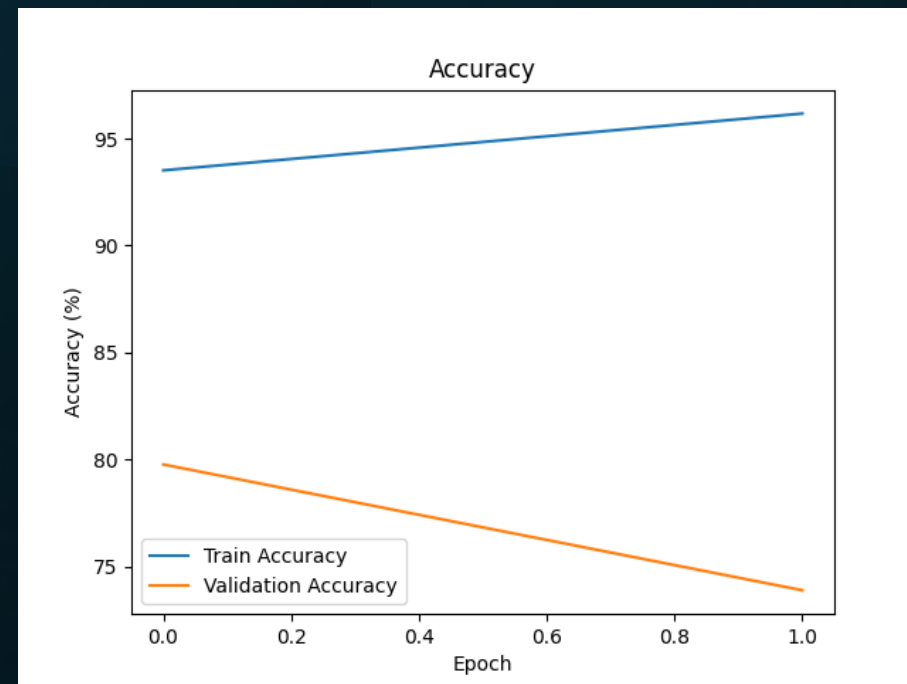
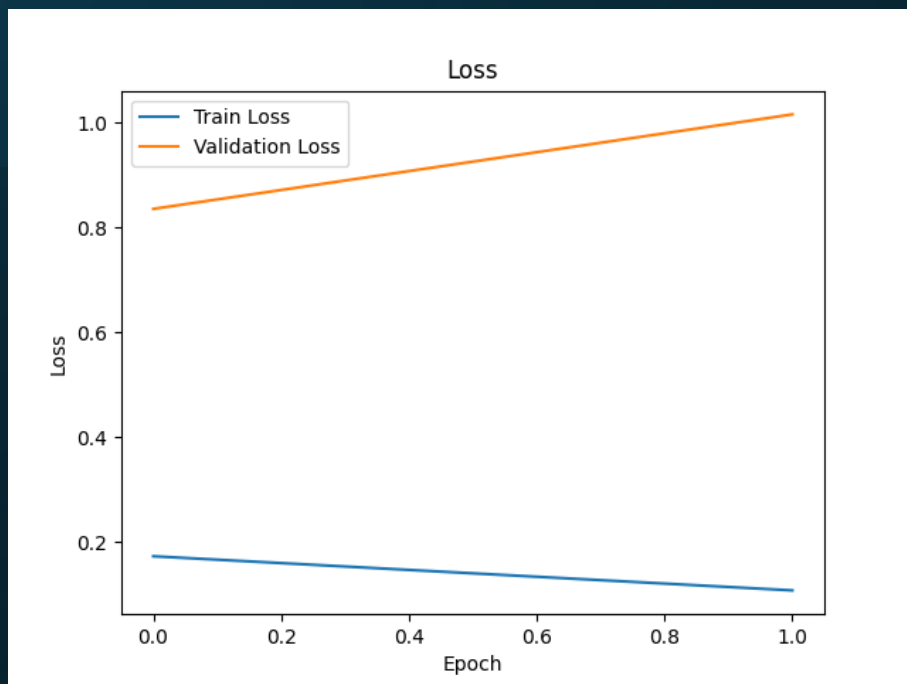
1 trening – 2 epoki



2 trening – po 4 epoche



3 trening – po 6 epoce



Przetrenowanie

Ulepszony preprocessing –
Normalize, RandomHorizontalFlip,
RandomRoation, ColorJitter

Zatrzymanie treningu po 4 epoche

Trenowanie modelu bezpiecznie (1, 2
epoki naraz)

Drugi model

Epoch 1

Train Loss: 0.7856

Train Accuracy: 70.24%

Validation Loss: 0.6789

Validation Accuracy: 76.87%

Epoch 2

Train Loss: 0.6335

Train Accuracy: 76.28%

Validation Loss: 0.6410

Validation Accuracy: 79.46%

Epoch 3

Train Loss: 0.5885

Train Accuracy: 78.07%

Validation Loss: 0.5893

Validation Accuracy: 81.14%

Epoch 4

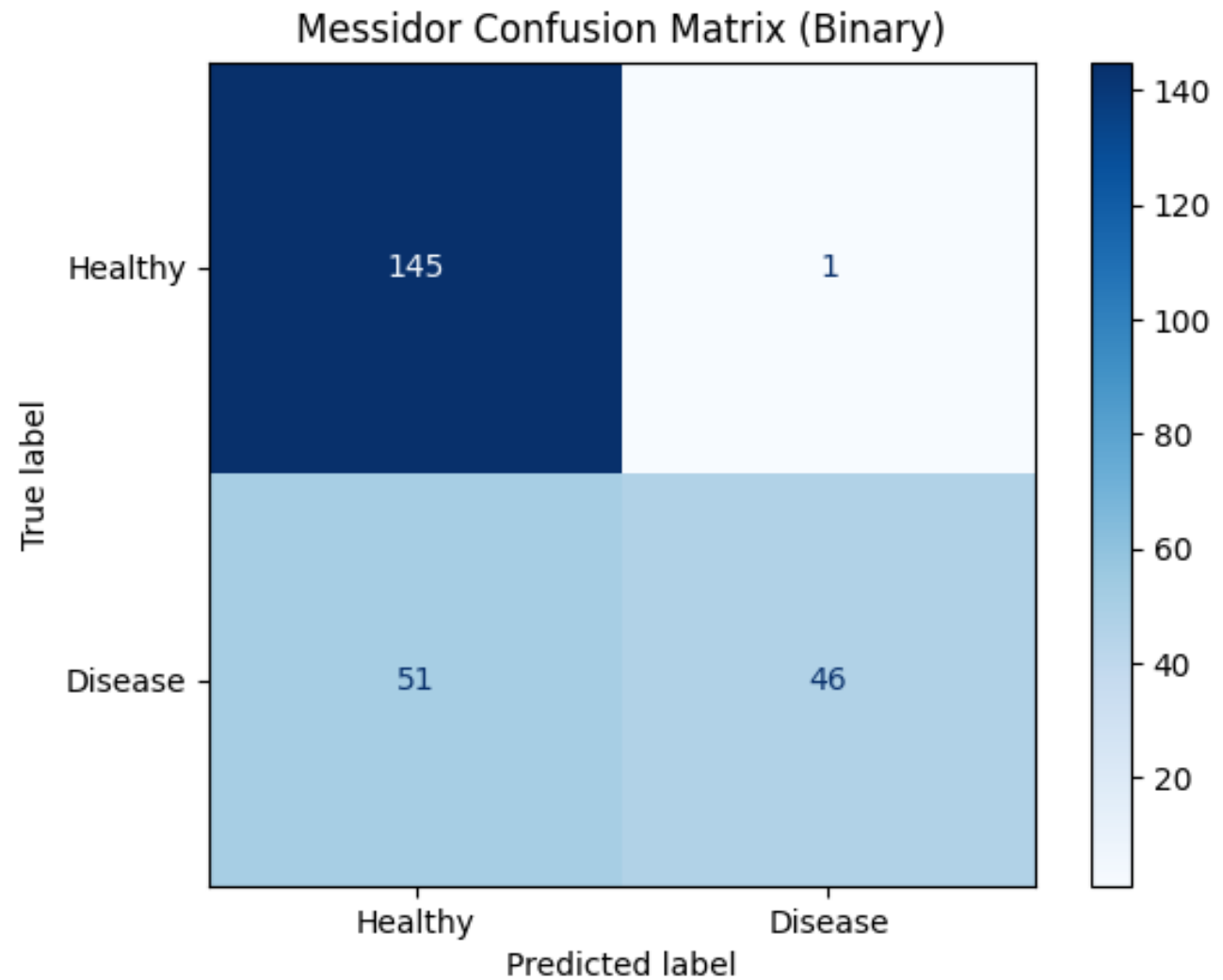
Train Loss: 0.5217

Train Accuracy: 79.94%

Validation Loss: 0.6316

Validation Accuracy: 80.38%

Confusion matrix po 4 epoche



Dodatkowy trening (1 epoka) + zmiana progu decyzyjnego

Epoch 5

Train Loss: 0.4702

Train Accuracy: 82.43%

Validation Loss: 0.6272

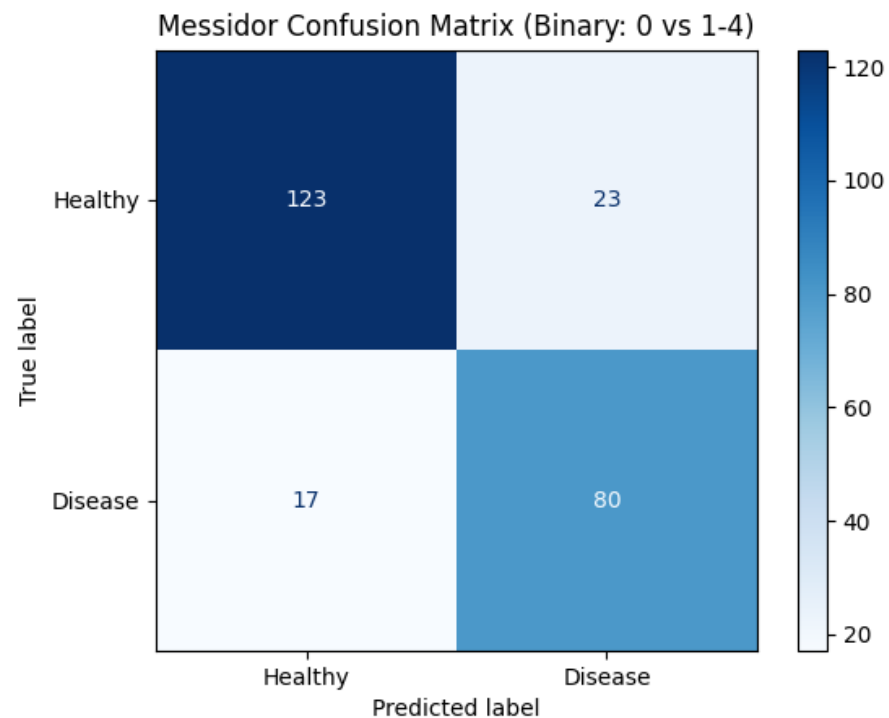
Validation Accuracy: 78.67%

Próg decyzyjny – 0,4:

Accuracy: 83,5%

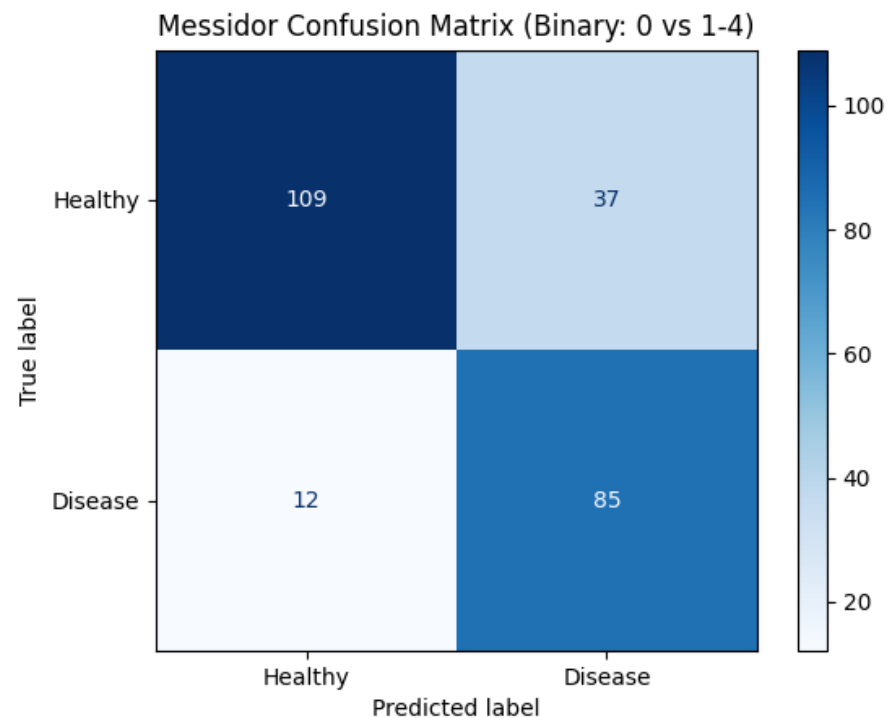
Recall: 82,5%

Precision: 77,7%



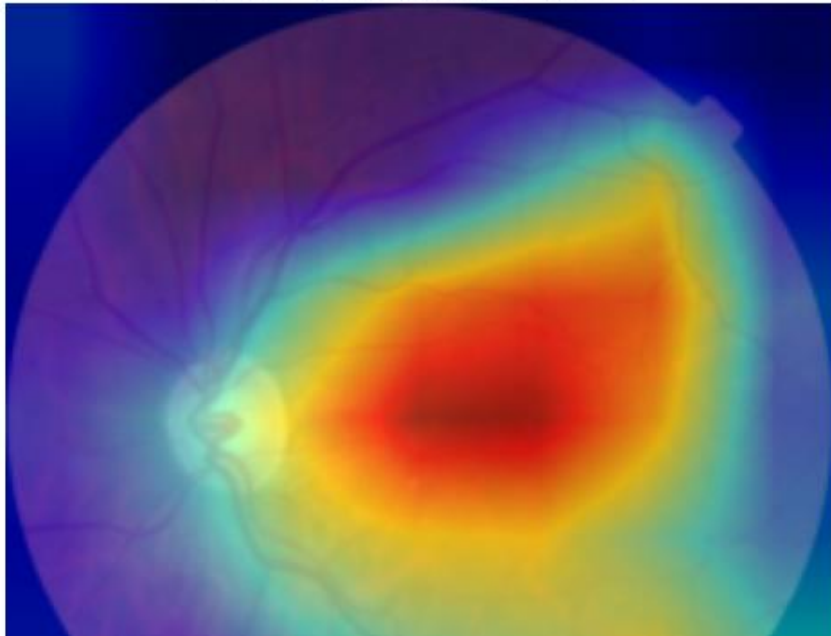
Porównanie progów decyzyjnego

- 0,2:
Accuracy: 66,3%
Precision: 54,5%
Recall: 92,8%
- 0,25:
Accuracy: 73,3%
Precision: 61,3%
Recall: 89,7%
- **0,3:**
Accuracy: 79,8%
Precision: 69,7%
Recall: 87,6%
- 0,4:
Accuracy: 83,5%
Recall: 82,5%
Precision: 77,7%

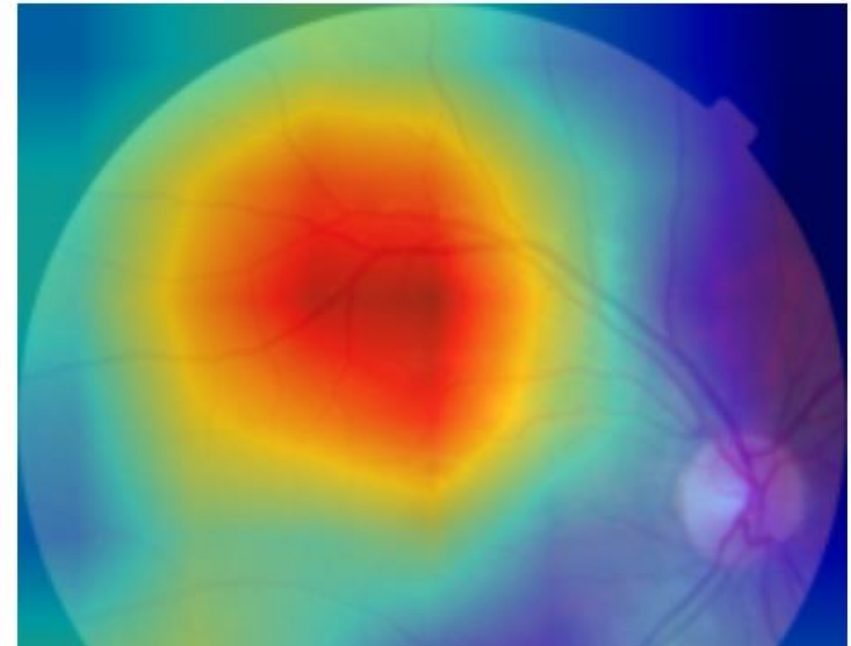


Grad-CAM (Gradient-weighted Class Activation Mapping)

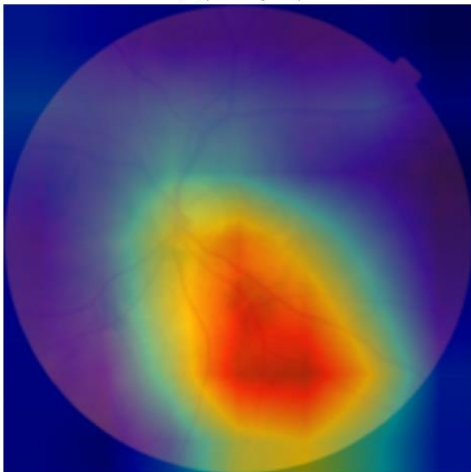
Pred: 0 (0) | Binary: 1 | True: 1



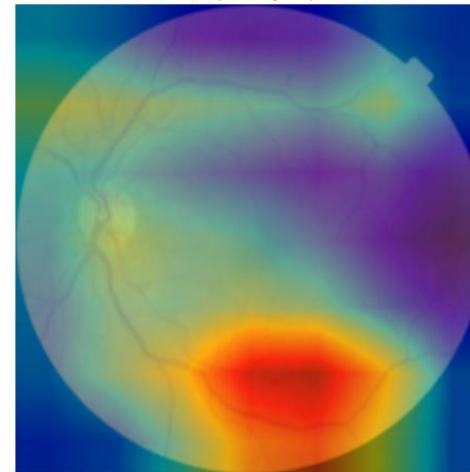
Pred: 0 (0) | Binary: 0 | True: 0



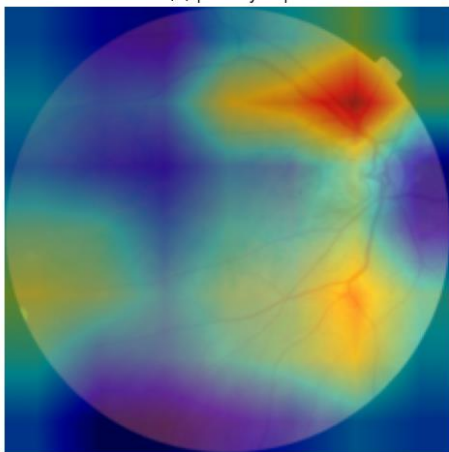
Pred: 2 (2) | Binary: 1 | True: 1



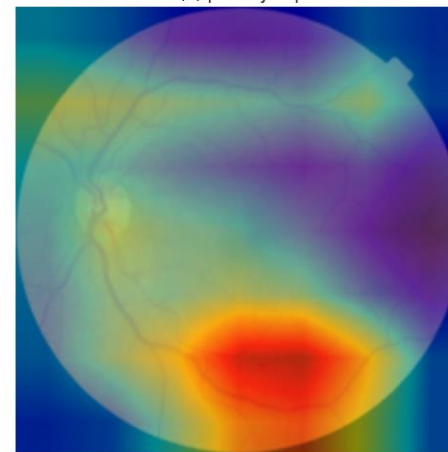
Pred: 2 (2) | Binary: 1 | True: 1



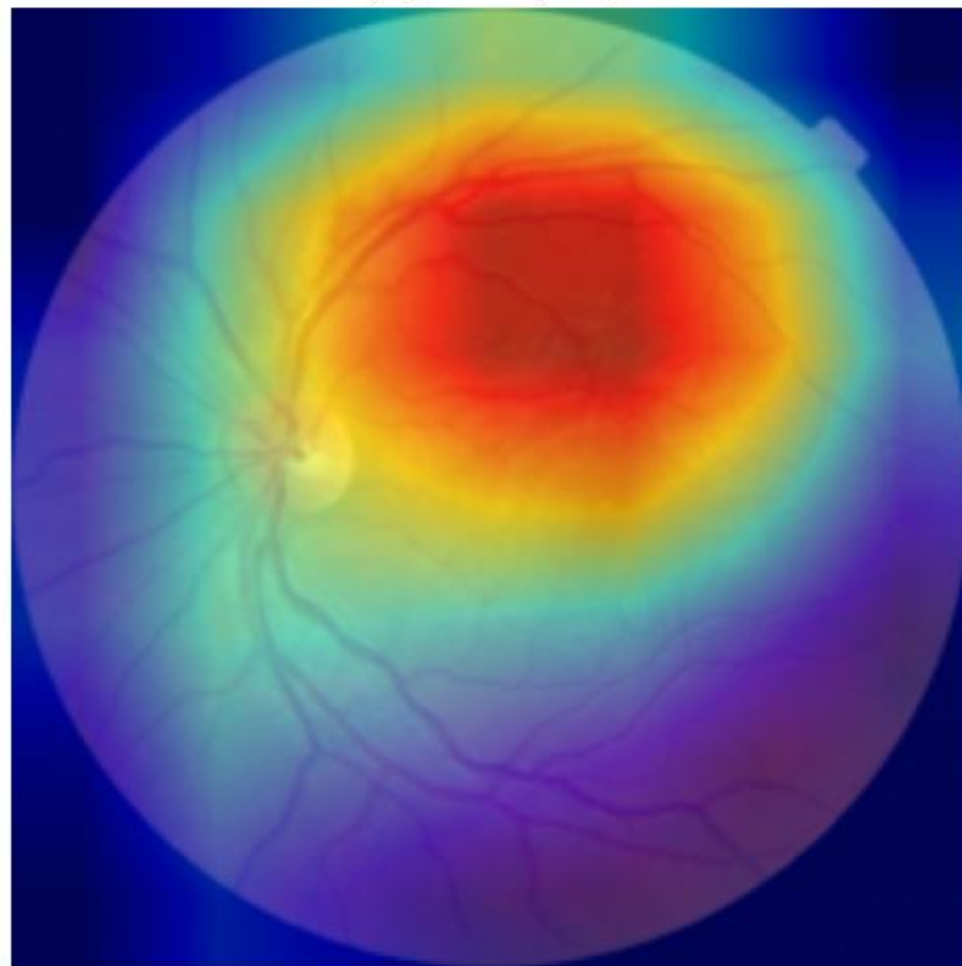
Pred: 2 (2) | Binary: 1 | True: 1



Pred: 2 (2) | Binary: 1 | True: 1



Pred: 1 (1) | Binary: 1 | True: 0



Kolejne kroki

EfficientNet, DenseNet

Trening na GPU

Trening i walidacja na większej liczbie zdjęć